

# YI-CHUN (CASEY) PAN

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## WORK EXPERIENCE

### MediaTek

03/2024 - 03/2026

Software Development Engineer

San Diego, USA

- Enhanced a multi-threaded DUT control system for 4 concurrent devices by optimizing mutex synchronization and shared memory handling, reducing RF calibration instrument ping time 60x (120s → 2s).
- Diagnosed I/O bottlenecks in production calibration pipelines and replaced MIPC file transfers with a usbfs kernel-level interface, reducing per-unit transfer time by 1.5s and improving factory throughput.
- Built and deployed a Jenkins CI/CD pipeline by provisioning a virtualized Jenkins master node, automating RF calibration across 4 concurrent DUTs with custom per-device test configurations.
- Owned reliability for customer-facing RF calibration software as technical DRI, leading root-cause analysis and resolution of 50+ critical regressions.
- Integrated LLM-based code review into Jenkins CI/CD pipelines, detecting code quality issues before human review and reducing PR iteration cycles.
- Refactored legacy C++ calibration modules using object-oriented design patterns, improving modularity, reducing technical debt, and cutting feature implementation time.
- Standardized Device Under Test (DUT) lab environments for internal and customer testing, creating SOPs that improved validation consistency and reproducibility.

### Zillow Group

05/2023 - 08/2023

Software Development Engineer Intern

Remote

- Built a reusable React.js micro-frontend archetype with integrated GitLab CI/CD, web configuration, and registration APIs, accelerating frontend service setup within the Shopper Platform.
- Implemented automated end-to-end Cypress testing in a scheduled GitLab CI/CD pipeline, catching regressions in the platform stub application before deployment.

### Amazon Web Services (AWS)

05/2021 - 04/2022

Cloud Support Intern

Taipei, Taiwan

- Built an AWS Lambda and DynamoDB-based on-call scheduling system with Slack API notifications, automating assignments and preventing scheduling conflicts for engineers.

## PROJECTS

### Fault-Tolerant Distributed Cluster for ML Workloads | CS 425 Distributed Systems

- Designed and implemented a fault-tolerant distributed system in C++ using TCP/UDP socket programming for coordinating ML training/inference workloads, featuring ring-based membership, SWIM-style failure detection, and a distributed job scheduling algorithm that maintained query processing balance within 20% across nodes.

### CUDA Kernel Optimization for CNN Inference | CS 483 Applied Parallel Programming

- Built and profiled CUDA convolution kernels for LeNet-5 (constant/shared memory, loop unrolling); tuned kernel parameters for 1.3x speedup, using Nsight Compute to diagnose why tiling and FP16 optimizations underperformed.

### Transfer Attack toward Model Training on Retail Dataset | CS 598 Transfer Learning

- Implemented and evaluated transfer-based adversarial attacks (PGD, FGSM, C&W) against ResNet-18, ResNet-50, and ViT models in a retail visual-perception setting.

### Real-Time Knowledge Management Chatroom | CS 546 Advanced Natural Language Processing

- Developed a React-based chatroom application integrating the T5 model for dynamic content structuring and summarization, enhancing user information management and cognitive flow.

## EDUCATION

### University of Illinois Urbana-Champaign

08/2022 – 12/2023

Master of Computer Science

Champaign, IL

### National Tsing Hua University

09/2017 – 01/2022

Bachelor of Science in Computer Science

Hsinchu, Taiwan

Minor in Quantitative Finance

## SKILLS

**Languages/Systems:** C/C++, CUDA, Multi-threading, Python, Java, JavaScript/TypeScript (React.js, Node.js), Linux

**Tools:** Jenkins, CMake, Docker, Git, REST API, SQL, Nsight Compute, Hugging Face Transformers, AWS (EC2, S3, Lambda), MongoDB, DynamoDB, Neo4j

**Spoken:** English (Proficient), Mandarin (Native)